

Gradient and Jacobian

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First, an example

Consider the bowl-shaped scalar field $f(x, y) = x^2 + y^2$. Its partial derivatives are $\partial f / \partial x = 2x$ and $\partial f / \partial y = 2y$, so the gradient is the vector field

$$\nabla f(x, y) = (2x, 2y).$$

Read aloud: “nabla f at x, y equals the vector two-x, two-y.”

At the point $(3, 4)$ this gives $\nabla f(3, 4) = (6, 8)$, a vector of length $\sqrt{6^2 + 8^2} = 10$ pointing radially outward from the origin. This is the direction of steepest ascent: moving an infinitesimal step along $(6, 8)$ increases f faster than any other direction of equal length, and the rate of that increase is 10. Negating it gives the direction of steepest descent—the move that gradient descent would take from $(3, 4)$. The rest of this lesson develops the gradient and its vector-valued cousin, the Jacobian, in full generality.

1 Setup and motivation

We work in finite-dimensional Euclidean space $\mathbb{R}^n$ with the standard inner product $\langle u, v \rangle = u^\top v$ and induced norm $\|v\| = \sqrt{\langle v, v \rangle}$. For scalar functions, the derivative captures a single rate of change; for vector-valued functions, it captures *all* pairwise rates between input and output coordinates, organized into a matrix.

2 Definitions

Definition 1 (Partial derivative). Let $U \subseteq \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}$. The *partial derivative* of f with respect to x_j at $x \in U$ is

$$\frac{\partial f}{\partial x_j}(x) = \lim_{h \rightarrow 0} \frac{f(x + h e_j) - f(x)}{h},$$

Read aloud: “partial f by partial x-j at x equals the limit, as h tends to zero, of f of x plus h times e-j minus f of x, all divided by h.”

where e_j is the j th standard basis vector, when this limit exists.

Definition 2 (Gradient). Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at x . The *gradient* of f at x is the column vector

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right)^\top \in \mathbb{R}^n.$$

Read aloud: “nabla f at x equals the column vector of partial f by partial x-1 through partial f by partial x-n, transposed, an element of R-to-the-n.”

Equivalently, $\nabla f(x)$ is the unique vector such that $D_v f(x) = \langle \nabla f(x), v \rangle$ for every $v \in \mathbb{R}^n$, where $D_v f(x) = \lim_{t \rightarrow 0} (f(x + tv) - f(x))/t$ is the directional derivative.

Definition 3 (Jacobian matrix). Let $F : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ with components $F = (F_1, \dots, F_m)$, differentiable at x . The *Jacobian matrix* of F at x is the $m \times n$ matrix

$$J_F(x) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1}(x) & \cdots & \frac{\partial F_1}{\partial x_n}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1}(x) & \cdots & \frac{\partial F_m}{\partial x_n}(x) \end{pmatrix}, \quad [J_F(x)]_{ij} = \frac{\partial F_i}{\partial x_j}(x).$$

Read aloud: “the i-j entry of the Jacobian J-F at x is partial F-i by partial x-j.”

$J_F(x)$ is the matrix of the total derivative $dF_x : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in standard bases.

Definition 4 (Jacobian determinant). If $m = n$, the *Jacobian determinant* of F at x is $\det J_F(x) \in \mathbb{R}$. It is the signed factor by which F locally scales n -dimensional volume at x : a small parallelepiped of volume V at x has image of volume $|\det J_F(x)| V + o(V)$.

3 The gradient and steepest ascent

Lemma 1 (Directional-derivative formula). *If f is differentiable at x , then for every $v \in \mathbb{R}^n$,*

$$D_v f(x) = \langle \nabla f(x), v \rangle.$$

Read aloud: “the directional derivative D_v of f at x equals the inner product of nabla f at x with v .”

Sketch. Differentiability at x means $f(x+h) = f(x) + \langle \nabla f(x), h \rangle + r(h)$ with $r(h)/\|h\| \rightarrow 0$. Setting $h = tv$ and dividing by t gives the claim in the limit $t \rightarrow 0$. $\square$

Theorem 2 (Direction of maximum directional derivative). *Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at $x \in U$, with $\nabla f(x) \neq 0$. Then among all unit vectors $v \in \mathbb{R}^n$ (i.e. $\|v\| = 1$), the directional derivative $D_v f(x)$ attains its maximum at*

$$v^* = \frac{\nabla f(x)}{\|\nabla f(x)\|},$$

Read aloud: “ v -star equals nabla f at x divided by the norm of nabla f at x .”

and the maximum value is $\|\nabla f(x)\|$.

Proof. By Lemma 1, $D_v f(x) = \langle \nabla f(x), v \rangle$. The Cauchy–Schwarz inequality states that for any $u, v \in \mathbb{R}^n$,

$$|\langle u, v \rangle| \leq \|u\| \|v\|,$$

Read aloud: “the absolute value of the inner product of u and v is at most the norm of u times the norm of v .”

with equality iff u and v are linearly dependent. Applying this with $u = \nabla f(x)$ and a unit vector v ,

$$D_v f(x) = \langle \nabla f(x), v \rangle \leq \|\nabla f(x)\| \|v\| = \|\nabla f(x)\|.$$

Read aloud: “ $D_v f$ at x equals the inner product of nabla f at x with v , which is at most the norm of nabla f at x times the norm of v , equal to the norm of nabla f at x .”

Equality holds iff v is a positive scalar multiple of $\nabla f(x)$; since $\|v\| = 1$ and $\nabla f(x) \neq 0$, the unique maximizer is $v^* = \nabla f(x)/\|\nabla f(x)\|$. Substituting back gives $D_{v^*} f(x) = \langle \nabla f(x), \nabla f(x)/\|\nabla f(x)\| \rangle = \|\nabla f(x)\|$. $\square$

Example 1. For $f(x, y) = x^2 + 3y^2$, $\nabla f(x, y) = (2x, 6y)$. At $(1, 1)$, the steepest-ascent direction is $(2, 6)/\sqrt{40} = (1, 3)/\sqrt{10}$ with rate $\sqrt{40}$, while $-\nabla f$ points toward the minimum at the origin.

4 The chain rule for vector-valued maps

Theorem 3 (Chain rule). *Let $G : \mathbb{R}^p \rightarrow \mathbb{R}^n$ be differentiable at x , and $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ differentiable at $y = G(x)$. Then $H = F \circ G$ is differentiable at x and*

$$J_H(x) = J_F(G(x)) \cdot J_G(x),$$

Read aloud: “the Jacobian of H at x equals the Jacobian of F at G of x , times the Jacobian of G at x .”

an $m \times p$ matrix obtained as the product of an $m \times n$ matrix and an $n \times p$ matrix.

This is the algebraic backbone of reverse-mode automatic differentiation: for a deep composition $F_L \circ F_{L-1} \circ \dots \circ F_1$, the overall Jacobian factors as a product of layer Jacobians, and the loss gradient propagates by successive vector–Jacobian products.

5 Worked example: $F(x, y) = (x^2 - y, e^{xy})$

The partials are $\partial F_1/\partial x = 2x$, $\partial F_1/\partial y = -1$, $\partial F_2/\partial x = ye^{xy}$, $\partial F_2/\partial y = xe^{xy}$. Hence

$$J_F(x, y) = \begin{pmatrix} 2x & -1 \\ ye^{xy} & xe^{xy} \end{pmatrix}, \quad J_F(1, 0) = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}, \quad \det J_F(1, 0) = 2.$$

Read aloud: “J-F at x, y is the two-by-two matrix with rows two- x , minus one and y -e-to-the- x - y , x -e-to-the- x - y ; at one-zero this becomes two, minus one, zero, one, with determinant two.”

By the inverse function theorem, F is a local diffeomorphism near $(1, 0)$, locally doubling area.